MORPHISMS OF LOCALLY RINGED TOPOI ON SMALL ETALE TOPOI COME FROM MORPHISMS OF SCHEMES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete construction: existence in the affine case, the global gluing, the compatibility of the 2-isomorphisms and uniqueness are all written out. The closing sentence 'Some details omitted' concerns routine compatibility checks in the gluing and is preserved verbatim in the counted span.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- The affine case of the theorem. Same chapter, Lemma lemma-morphism-ringed-etale-topoi-affines, whose proof is adjusted to the global setting. Supplies the local existence statement that is glued.
- Rigidity of morphisms of locally ringed topoi. Same chapter, Lemma lemma-2-morphism. Makes the local morphisms agree on overlaps and the 2-isomorphisms compatible.
- Localization morphisms of locally ringed topoi attached to open immersions. Same chapter, Lemma lemma-morphism-locally-ringed. Supplies the morphisms along which the local statements are composed.

Source reference. Original lemma 04I6: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 5347–5347. Navigation pointer invoked at proof line 5576; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 04IJ: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 5227–5227. Navigation pointer invoked at proof

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line 5580; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 04I5: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 5178–5178. Navigation pointer invoked at proof line 5592; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 04IG: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 5603–5603. Navigation pointer invoked at proof line 5603; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 04LW: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 5298–5298. Navigation pointer invoked at proof line 5628; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original definition 003H: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 5647–5647. Navigation pointer invoked at proof line 5647; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original section 04I9: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 5649–5649. Navigation pointer invoked at proof line 5649; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 04HV: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 3923–3923. Navigation pointer invoked at proof line 5667; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 04IH: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 5670–5670. Navigation pointer invoked at proof line 5670; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Theorem 0.1. *Let X, Y be schemes. Let*

$$(g, g^\#) : (Sh(X_{étale}), \mathcal{O}_X) \longrightarrow (Sh(Y_{étale}), \mathcal{O}_Y)$$

be a morphism of locally ringed topoi. Then there exists a unique morphism of schemes $f : X \to Y$ such that $(g, g^\#)$ is isomorphic to $(f_{small}, f^\sharp_{small})$. In other words, the construction

$$Sch \longrightarrow Locally\ ringed\ topoi, \quad X \longrightarrow (X_{étale}, \mathcal{O}_X)$$

is fully faithful (morphisms up to 2-isomorphisms on the right hand side).

Proof. You can prove this theorem by carefully adjusting the arguments of the proof of Lemma 04I6 to the global setting. However, we want to indicate how we can glue the result of that lemma to get a global morphism due to the rigidity provided by the result of Lemma 04IJ. Unfortunately, this is a bit messy.

Let us prove existence when Y is affine. In this case choose an affine open covering $X = \bigcup U_i$. For each i the inclusion morphism $j_i : U_i \to X$ induces a morphism of locally ringed topoi $(j_{i,small}, j_{i,small}^\sharp) : (Sh(U_{i,étale}), \mathcal{O}_{U_i}) \to (Sh(X_{étale}), \mathcal{O}_X)$ by Lemma 04I5. We can compose this with $(g, g^\sharp)$ to obtain a morphism of locally ringed topoi

$$(g, g^\sharp) \circ (j_{i,small}, j_{i,small}^\sharp) : (Sh(U_{i,étale}), \mathcal{O}_{U_i}) \to (Sh(Y_{étale}), \mathcal{O}_Y)$$

see Modules on Sites, Lemma 04IG. By Lemma 04I6 there exists a unique morphism of schemes $f_i : U_i \to Y$ and a 2-isomorphism

$$t_i : (f_{i,small}, f_{i,small}^\sharp) \longrightarrow (g, g^\sharp) \circ (j_{i,small}, j_{i,small}^\sharp).$$

Set $U_{i,i'} = U_i \cap U_{i'}$, and denote $j_{i,i'} : U_{i,i'} \to U_i$ the inclusion morphism. Since we have $j_i \circ j_{i,i'} = j_{i'} \circ j_{i',i}$ we see that

$$\begin{aligned}(g, g^\sharp) \circ (j_{i,small}, j_{i,small}^\sharp) \circ (j_{i,i',small}, j_{i,i',small}^\sharp) = \\ (g, g^\sharp) \circ (j_{i',small}, j_{i',small}^\sharp) \circ (j_{i',i,small}, j_{i',i,small}^\sharp)\end{aligned}$$

Hence by uniqueness (see Lemma 04LW) we conclude that $f_i \circ j_{i,i'} = f_{i'} \circ j_{i',i}$, in other words the morphisms of schemes $f_i = f \circ j_i$ are the restrictions of a global morphism of schemes $f : X \to Y$. Consider the diagram of 2-isomorphisms (where we drop the components $\sharp$ to ease the notation)

$$\begin{array}{ccc} g \circ j_{i,small} \circ j_{i,i',small} & \xrightarrow{t_i \star \mathrm{id}_{j_{i,i',small}}} & f_{small} \circ j_{i,small} \circ j_{i,i',small} \\ \| & & \| \\ g \circ j_{i',small} \circ j_{i',i,small} & \xrightarrow{t_{i'} \star \mathrm{id}_{j_{i',i,small}}} & f_{small} \circ j_{i',small} \circ j_{i',i,small} \end{array}$$

The notation $\star$ indicates horizontal composition, see Categories, Definition 003H in general and Sites, Section 04I9 for our particular case. By the result of Lemma 04IJ this diagram commutes. Hence for any sheaf $\mathcal{G}$ on $Y_{étale}$ the isomorphisms $t_i : f_{small}^{-1}\mathcal{G}|_{U_i} \to g^{-1}\mathcal{G}|_{U_i}$ agree over $U_{i,i'}$ and we obtain a global isomorphism $t : f_{small}^{-1}\mathcal{G} \to g^{-1}\mathcal{G}$. It is clear that this isomorphism is functorial in $\mathcal{G}$ and is compatible with the maps $f_{small}^\sharp$ and $g^\sharp$ (because it is compatible with these maps locally). This proves the theorem in case Y is affine.

In the general case, let $V \subset Y$ be an affine open. Then h_V is a subsheaf of the final sheaf $*$ on $Y_{étale}$. As g is exact we see that $g^{-1}h_V$ is a subsheaf of the final sheaf on $X_{étale}$. Hence by Lemma 04HV there exists an open subscheme $W \subset X$ such that $g^{-1}h_V = h_W$. By Modules on Sites, Lemma 04IH there exists a commutative diagram of morphisms of locally ringed topoi

$$\begin{array}{ccc} (Sh(W_{étale}), \mathcal{O}_W) & \longrightarrow & (Sh(X_{étale}), \mathcal{O}_X) \\ \downarrow{\scriptstyle g'} & & \downarrow{\scriptstyle g} \\ (Sh(V_{étale}), \mathcal{O}_V) & \longrightarrow & (Sh(Y_{étale}), \mathcal{O}_Y) \end{array}$$

where the horizontal arrows are the localization morphisms (induced by the inclusion morphisms $V \to Y$ and $W \to X$) and where g' is induced from g. By the result of the preceding paragraph we obtain a morphism of schemes $f' : W \to V$ and a 2-isomorphism $t : (f'_{small}, (f'_{small})^\sharp) \to (g', (g')^\sharp)$. Exactly as before these

morphisms f' (for varying affine opens $V \subset Y$) agree on overlaps by uniqueness, so we get a morphism $f : X \to Y$. Moreover, the 2-isomorphisms t are compatible on overlaps by Lemma 04IJ again and we obtain a global 2-isomorphism $(f_{small}, (f_{small})^\sharp) \to (g, (g)^\sharp)$. as desired. Some details omitted. $\square$